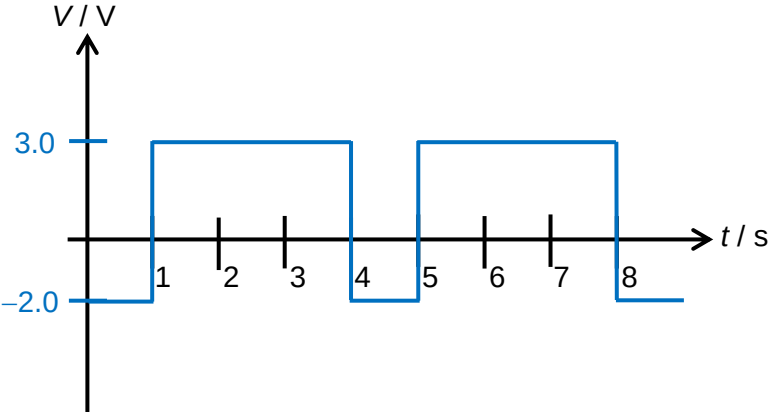


|   |          |                                                                                                                                                                                                                                                                                                                                              |                                         |
|---|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 5 | (a)(i)   | $\langle V^2 \rangle = [(3^2 \times 3) + (6^2 \times 1)] / 4$ $= 15.75 \text{ V}^2$ $V_{\text{rms}} = (15.75)^{0.5}$ $= 4.0 \text{ V}$                                                                                                                                                                                                       | <b>C1</b><br><b>A1</b>                  |
|   | (a)(ii)  |  <p>Shape (not rectified) with correct period – 1m</p> <p>Correct values either (3.0 V, -2.0 V) Or (-6.0 V, 1.0 V) – 1m</p>                                                                                                                                | <b>B1</b><br><b>B1</b>                  |
|   | (b)(i)   | Turns ratio = $4.50 : 500 = 9 : 1000 = 0.009$<br>(allow for $500 : 4.50 = 111$ )                                                                                                                                                                                                                                                             | <b>A1</b>                               |
|   | (b)(ii)  | transmission line resistance = $6.44 \times 10^5 \times 4.50 \times 10^{-4} = 289.8 \, \Omega$<br>Given the voltage is stepped up to 500 kV at the output of the transformer,<br>The current flowing in the tx line = $5 \times 10^6 / 500 \times 10^3 = 10 \text{ A}$<br>Power loss = $10^2(289.8) = \underline{2.9 \times 10^4 \text{ W}}$ | <b>C1</b><br><br><b>C1</b><br><b>A1</b> |
|   | (b)(iii) | By stepping up the voltage, the <u>current in the transmission line will be lowered</u> and hence the amount of <u>power loss in the transmission line will be lowered.</u>                                                                                                                                                                  | <b>A1</b>                               |

|   |        |                                                                         |    |
|---|--------|-------------------------------------------------------------------------|----|
| 6 | (a)(i) | The spread of the electron beam to form bright rings due to diffraction | B1 |
|---|--------|-------------------------------------------------------------------------|----|